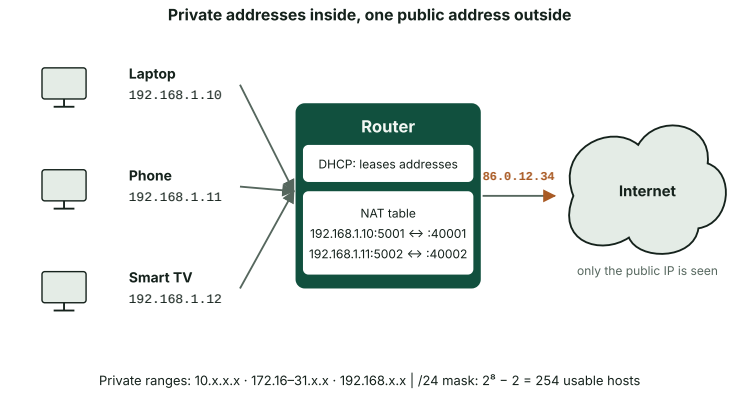
IP addressing, switching and routing

IS11 FACT FILE

Specification: students should be able to

- Describe IPv4 and IPv6 addressing, public and private addresses and the purpose of subnetting
- Explain the roles of DHCP and NAT
- Compare packet switching and circuit switching
- Describe how routing works

Key words: IP address, IPv4, IPv6, private address, public address, subnet mask, DHCP, NAT, packet switching, circuit switching, routing table, hop



A home network where DHCP gives devices private IP addresses and the router uses NAT to share one public IP address with the internet

IP addressing

An IP address is a logical address identifying a device on a network. IPv4 uses 32 bits written as four denary octets, giving about 4.3 billion addresses, which proved insufficient. IPv6 uses 128 bits written in hexadecimal, providing a vast address space. Private address ranges such as 192.168.0.0/16 are used inside LANs and not routed on the internet, while public addresses are unique worldwide. A subnet mask separates the network part of an address from the host part.

DHCP, NAT and subnets

DHCP automatically allocates an IP address, subnet mask, default gateway and DNS server to each device for a lease period, avoiding manual errors and duplicate addresses. NAT allows many devices with private addresses to share one public IPv4 address by rewriting addresses at the router and tracking each connection. Subnetting divides a network into smaller logical networks, reducing broadcast traffic, improving security and simplifying management.

Switching and routing

In circuit switching a dedicated path is reserved for the whole communication, giving a constant rate but wasting capacity during silence. In packet switching data is broken into packets with headers containing addresses and sequence numbers; packets travel independently, possibly by different routes, and are reassembled at the destination. Routers use routing tables to forward each packet to the next hop, so traffic can avoid congested or failed links and capacity is shared efficiently.

Key terms from the mark schemes

IP address
A logical numerical address that identifies a device on a network so data can be routed to it.

DHCP
Dynamic Host Configuration Protocol: a service that automatically assigns IP addresses and network settings to devices.

Packet switching
A method of transmission in which data is split into packets that travel independently across shared links and are reassembled at the destination.

Subnet mask
A value that identifies which bits of an IP address form the network part and which form the host part.

Network address translation (NAT)
A router technique that maps private IP addresses to a public IP address so many devices can share one public address.

Circuit switching
A method of transmission in which a dedicated path is set up and reserved between sender and receiver for the whole communication.

Exam focus

Calculation questions may ask for the number of addresses or usable hosts: find the host bits (32 minus the prefix length), calculate 2 to that power and subtract 2 for the network and broadcast addresses. Show your working for method marks. For comparison questions use matching criteria for both methods. When explaining packet switching, mention splitting into packets, headers with destination address and sequence number, independent routes, and reassembly – each is a creditable point.

? Check yourself

1. State the number of bits in an IPv4 address and in an IPv6 address. [2]
2. Explain why IPv6 was introduced. [2]
3. A network uses the subnet mask 255.255.255.192 (/26). Calculate the number of usable host addresses. Show your working. [3]

Answers: 1. IPv4: 32 bits (1) / IPv6: 128 bits (1). 2. IPv4 provides only about 4.3 billion (2^32) addresses, which have run out (1) / because of the huge growth in internet-connected devices such as smartphones and IoT devices; IPv6's 128-bit addresses give about 3.4 × 10^38 addresses (1). 3. Host bits = 32 − 26 = 6 (1) / 2^6 = 64 addresses (1) / subtract network and broadcast addresses: 64 − 2 = 62 usable hosts (1).